Causes of Hospitalization Linked to Short-Term Air Pollution / Weather Exposure

Based on Dr. Francesca Dominici's peer-reviewed publications (last 15 years, 2011–2026)

This list covers papers on her CV that examine **short-term** (daily/weekly, case-crossover, time-series, or single-event) associations between air pollution (PM2.5, ozone, CO, wildfire smoke, PM constituents) or weather/climate variables (heat, cold, snowstorms, drought, hurricanes/tropical cyclones, volcanic SO2) and hospital admissions or ED visits. Long-term/chronic cohort exposure papers (e.g., mortality-only or multi-year exposure studies) are excluded unless noted.

ICD-9-CM and ICD-10-CM codes are provided at the category level typically used in these studies (claims-based Medicare/Medicaid hospitalization research). Where a paper does not specify exact codes, standard epidemiologic definitions for that condition are given.

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1. From Dr. Dominici's Publications

Cause of Hospitalization / ED Visit	ICD-9-CM	ICD-10-CM	Representative Publication(s)
Cardiovascular disease (all-cause)	390–459	I00–I99	Dominici et al. 2006 JAMA (#57); Peng et al. 2008 JAMA (#78); Bravo et al. 2016 (#172); Chen et al. 2021 (#245, #253); Sun et al. 2024 BMJ (#304); Yazdi et al. 2025 (#323)
Respiratory disease (all-cause)	460–519	J00–J99	Dominici et al. 2006 JAMA (#57); Peng et al. 2008 JAMA (#78); Bell et al. 2009 (#89, #94); Bravo et al. 2016 (#172); Anderson et al. 2013 (#132)
Ischemic heart disease	410–414	I20–I25	Dominici et al. 2006 JAMA (#57)
Myocardial infarction (acute)	410	I21–I22	Barr et al. 2012 (#125, smoking bans); Schwartz et al. 2023 (#284)
Congestive heart failure	428	I50	Symons et al. 2006 (#61); Dominici et al. 2006 JAMA (#57)
Cardiac arrhythmia	427	I47–I49	Dominici et al. 2006 JAMA (#57, cardiovascular admissions category)
Cerebrovascular disease / stroke	430–438	I60–I69	Dominici et al. 2006 JAMA (#57, cardiovascular category)
Carbon monoxide–associated cardiovascular admissions	410–414, 427–428, 786.5 (chest pain)	I20–I25, I47–I50, R07.9	Bell et al. 2009, *Circulation* (#92)
Chronic obstructive pulmonary disease (COPD)	490–492, 494–496	J40–J44	Peng et al. 2008 JAMA (#78); Bell et al. 2009 (#89, #94)
Asthma	493	J45	Wei et al. 2022, *AJRCCM* (#271); Bell et al. 2009 (#89, #94)
Pneumonia / acute	480–487	J09–J18	Peng et al. 2008 JAMA (#78,

respiratory infection			respiratory category)
Acute respiratory distress syndrome (ARDS)	518.82	J80	Rhee et al. 2019 (#202); Rhee et al. 2020 (#229)
Cardiorespiratory hospitalizations (combined CVD + respiratory)	390–519	I00–I99, J00–J99	Yan et al. 2021 (#242, tropical cyclones); Thomas et al. 2020/2022 (#220, #268)
Diabetes-related hospitalization/ exacerbation	250	E10–E14	Zanobetti et al. 2014 (#148)
Neurological disorders (general)	320–359	G00–G99	Zanobetti et al. 2014 (#148)
Chronic debilitating brain disorders	290–294, 320–359	F01–F09, G00–G99	Yitshak-Sade et al. 2020 (#238)
Dementia / Alzheimer's disease	290.0, 331.0	F01–F03, G30	Lee et al. 2019 (#212); Qin et al. 2024 (#311); Dong et al. 2025 (#320); Spoto et al. 2025 (#339)
Acute kidney injury / kidney disease	584, 585	N17, N18	Lee et al. 2023, *EHP* (#282)
Psychiatric hospital admissions	290–319	F01–F99	Qiu et al. 2022, *Lancet Planetary Health* (#270)
Alcohol- and substance-related disorders	291, 303–305	F10–F19	Parks et al. 2023, *Communications Medicine* (#293)
Psychiatric hospitalizations (boom–bust economic/oil-gas exposure context)	290–319	F01–F99	Willis et al. 2025 (#331)
Interpersonal violence–related visits (assault)	E960–E969	X92–Y09	Parks et al. 2026 (#341)
Self-inflicted violence–related visits	E950–E959	X71–X83	Parks et al. 2026 (#341)
Heat stroke / heat-related illness	992.0 (heat stroke), 992.0–992.9 (heat illness	T67.0XXA (heatstroke), T67.0–T67.9	Wang et al. 2016 (#171); Bobb et al. 2014, JAMA (#153); Hopp et al. 2018 (#190)

	992.9 (heat illness spectrum)	T67.0–T67.9	Hopp et al. 2018 (#196)
Heat-related respiratory hospitalization	460–519	J00–J99	Anderson et al. 2013, *AJRCCM* (#132)
Cause-specific hospitalizations related to extreme heat (renal, respiratory, CVD, diabetes, etc., multi-cause)	Multiple (250, 390–519, 584–586)	Multiple (E10–E14, I00–I99, J00–J99, N17–N19)	Bobb (Obermeyer) et al. 2014, JAMA "Cause-Specific Risk of Hospital Admission Related to Extreme Heat" (#153)
Wildfire smoke–associated hospital admissions (respiratory/cardiovascular)	460–519, 390–459	J00–J99, I00–I99	Liu et al. 2016 (#173); Liu et al. 2017 (#175)
Snowstorm-associated, cause-specific hospital admissions	Varies by cause (injury, cardiovascular, respiratory)	Varies (S00–T88, I00–I99, J00–J99)	Bobb et al. 2016, *AJE* "Time-course of Cause-Specific Hospital Admissions during Snowstorms" (#177)
Drought-associated hospital admissions/mortality (older adults)	Not cause-specific in abstract; general morbidity	—	Berman et al. 2017, *Lancet Planetary Health* (#181)
Tropical cyclone–associated hospitalization (older adults, general and cardiorespiratory)	390–519	I00–I99, J00–J99	Parks et al. 2021, *Nature Communications* (#240); Yan et al. 2021 (#242); Burrows et al. 2023 (#281)
Hurricane-associated ED visits (by age and cause)	Multiple, cause-stratified	Multiple	Weinberger et al. 2021, *AJE* (Hurricane Sandy) (#244)
Volcanic SO2 plume–associated respiratory morbidity	460–519 (esp. 493 asthma, 490–496 COPD)	J00–J99 (esp. J45, J40–J44)	Carlsen et al. 2021 (#250, #252)
Emergency hospital admissions for CVD and ambient CO	See CVD row above	See CVD row above	Bell et al. 2009, *Circulation* (#92)
Hospital admissions and PM2.5 chemical composition (cardiorespiratory)	390–519	I00–I99, J00–J99	Bell et al. 2009 (#89, #94); Bell et al. 2014 (#136)
Emergency hospital	390–519	I00–I99, J00–	Kioumourtzoglou et al. 2013

admissions, primary organic particles (elderly, cardiorespiratory)		J99	(#134)
Non-respiratory infection hospital admissions (long-term PM2.5, included for completeness)	001–139, 320–326, 590, etc. (infection-specific)	A00–B99 and site-specific infection codes	Feng et al. 2024, *Nature Communications* (#303/#309)
Second cardiovascular event hospital admissions	410–414, 428, 430–438	I20–I25, I50, I60–I69	Yazdi et al. 2025 (#323)

Note: Several of these papers (e.g., #57, #78, #92, #89/#94, #136) report cardiovascular and respiratory hospital admissions as broad, pre-specified diagnostic categories (often built from CMS/Medicare claims) rather than single ICD codes — the ranges above reflect the standard groupings used in that literature (e.g., NMMAPS/Medicare cardiovascular and respiratory admission definitions).

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2. Additional ICD-9/ICD-10 Codes Associated with Short-Term Air Pollution or Weather (Broader Environmental Epidemiology Literature — Not Necessarily in the CV)

These are causes of hospitalization commonly linked to short-term air pollution or weather exposure in the wider field, even though they don't appear explicitly in the listed publications.

Cause	ICD-9-CM	ICD-10-CM	Typical Exposure Studied
Hypertensive crisis / hypertension	401–405	I10–I16	Air pollution, heat
Atrial fibrillation (specific arrhythmia)	427.31	I48.0–I48.9	Air pollution, temperature
Cardiac arrest	427.5	I46.2, I46.8, I46.9	Ozone, PM2.5, extreme heat/cold
Chronic bronchitis	491	J41–J42	Air pollution
Acute bronchitis	466	J20–J21	Air pollution, cold weather
Upper respiratory tract infection	460–465	J00–J06	Cold weather, air pollution
Otitis media	381–382	H65–H67	Air pollution (pediatric)
Conjunctivitis / eye irritation	372.0	H10	Ozone, particulate matter, wildfire smoke
Allergic rhinitis	477	J30	Pollen, air pollution interaction
Dehydration	276.51	E86.0	Extreme heat
Heat exhaustion / heat cramps	992.3, 992.5–992.7	T67.3, T67.2, T67.6–T67.8	Extreme heat
Hypothermia	991.6	T68	Extreme cold
Frostbite	991.0–991.3	T33–T34	Extreme cold
Cold exposure–related cardiovascular events	410–414, 427–428	I20–I25, I47–I50	Cold weather
Renal colic / kidney stones	592	N20	Extreme heat, dehydration
Sepsis / urinary tract infection (heat-related renal stress)	038, 599.0	A41, N39.0	Extreme heat
Rhabdomyolysis	728.88	M62.82	Extreme heat
Falls / weather-related injury (ice, snow)	800–959 (injury codes), E880–E888	S00–T88, W00–W19	Snow/ice storms
Carbon monoxide poisoning	986	T58	Cold weather (heating

			equipment use), power outages
Preterm birth / adverse birth outcomes	644.2, 765	O60, P07	Air pollution (PM2.5, NO2), heat
Low birth weight	764	P05	Air pollution
Vector-borne disease hospitalizations (e.g., dengue, West Nile virus)	061, 066.4	A90–A91, A92.3	Temperature/climate variability
Gastroenteritis / diarrheal disease	558.9, 009	A09, K52.9	Extreme heat, flooding
Mental health crisis visits (general, beyond psychiatric admission category)	300–316	F40–F48, F90–F98	Heat, air pollution
Suicide attempt–related ED visits	E950–E959	X71–X83, T14.91	Heat
Anxiety/mood disorder exacerbation	296, 300	F30–F39, F40–F41	Air pollution, heat
Wildfire smoke–specific asthma/ COPD exacerbation	493, 491–492, 496	J45, J41–J44	Wildfire PM2.5
All-cause emergency department visits (non-specific)	—	—	Heat, air pollution (used as sensitivity outcome)

Notes on Interpretation

1. **ICD-9-CM vs. ICD-10-CM:** U.S. claims data (Medicare/Medicaid) transitioned from ICD-9-CM to ICD-10-CM on October 1, 2015. Papers using data before this date use ICD-9; papers using more recent data use ICD-10 or a crosswalk.
2. **"Cardiovascular" and "respiratory" admissions** in air pollution epidemiology are frequently defined using standardized code groupings from the NMMAPS/Medicare literature (often built off the AHRQ Clinical Classifications Software or the ICD-9 ranges 390–459 and 460–519, respectively) rather than single diagnosis codes — this is the convention in most of Dr. Dominici's cardiorespiratory papers.
3. Where a paper's exact ICD code list was not published in the abstract/title, the codes shown reflect standard definitions used in equivalent published methods (e.g., Medicare claims-based hospitalization studies).